

DELARAM HEYDARIAN

Curriculum Vitae

Department of Industrial Engineering and Management Sciences | Northwestern University | Evanston, IL
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[https://scholar.google.com/citations?user=qwn96hAAAAAJ&hl=en]

EDUCATION

2021–Present	Ph.D. in Operation research, Northwestern University Evanston, IL, USA Research area: Operations Management, Service Operations, Stochastic Systems Advisors: Prof. Achal Bassamboo and Prof. Seyed Iravani Expected graduation: 2027
[2019-2021]	M.S. in Management Science, Miami Herbert Business School, University of Miami Coral Gables, FL, USA
[2014-2017]	M.S. in Industrial Engineering, University of Tehran Tehran, Iran
[2008-2013]	B.S. in Mathematics, Sharif University of Technology Tehran, Iran

RESEARCH INTERESTS AND METHODS

Interests: Operations Management • Operation Research • Service Operations • Queueing Systems • Customer Routing and Choice • Congestion Externalities • Heterogeneous Service Systems • Decentralized Decision Making

Methods: Queueing Theory • Stochastic Modeling • Applied Probability • Equilibrium Analysis • Optimization • Simulation • Numerical Analysis

RESEARCH

Checkout Externalities: A Slow Server Problem with Two Classes (*Job Market Paper / Working Paper*)

Studies customer routing and congestion externalities in service systems with heterogeneous servers and heterogeneous customer classes. The model is motivated by checkout systems in which customers choose between a fast traditional checkout and a slower self-checkout system. I characterize decentralized customer-routing behavior using Wardrop equilibrium and compare it with the social planner's optimal routing policy. The project combines queueing-theoretic analysis, equilibrium modeling, optimization, and simulation.

with Prof. Achal Bassamboo and Prof. Martin Lariviere

Customer Routing in Heterogeneous Service Systems (*Working Research Project*)

Studies routing decisions when heterogeneous customers choose between servers with different service capabilities, with emphasis on equilibrium routing, socially optimal allocation, service-time variability, and structural properties of mixing and pure-routing strategies. Methods include M/G/1 queueing models, Pollaczek–Khinchine analysis, stochastic modeling, Wardrop equilibrium, optimization, and computational experiments.

with Prof. Achal Bassamboo and Prof. Martin Lariviere, and Prof. Seyed MR Iravani

Additional Research

- [Paper title] — [status, coauthors, short description]
- [Paper title] — [status, coauthors, short description]
- [Paper title] — [status, coauthors, short description]

TEACHING

IEMS 381 — [Supply Chain Management]

Teaching Assistant and Lab Instructor

Northwestern University | Fall 2024, Fall 2025, Fall 2026

Held office hours and provided support to students on course concepts and assignments.

Graded homework, projects, and other course assessments.

Led and taught lab sessions, guiding students through supply chain management exercises and applications.

EMS 382 — [Operation Management]

Teaching Assistant

Northwestern University | Winter 2023, Winter 2024, and Winter 2025

Held office hours and provided support to students on course concepts and assignments.

Graded homework, projects, and other course assessments.

IEMS 383 — [Service Management]

Teaching Assistant, Northwestern University | [Spring 2023, Spring 2024, and Spring 2025]

Teaching responsibilities have included supporting undergraduate instruction, holding office hours, answering student questions, grading assignments and projects, and assisting with examinations.

PRESENTATIONS

Checkout Externalities: A Slow Server Problem with Two Classes

- INFORMS Annual Meeting 2026 — [location/session]
- POMS Annual Conference 2026 — Reno, NV
- University of Chicago Workshop

RESEARCH AND COMPUTATIONAL SKILLS

Mathematical and research methods: Queueing Theory, Stochastic Processes, Applied Probability, Markov Processes, Optimization, Equilibrium Analysis, Simulation, Statistical Analysis, Numerical Optimization

Programming: R (advanced research and simulation use), Mathematica / Wolfram Language, Python, LaTeX

PROFESSIONAL DEVELOPMENT AND CONFERENCES

- INFORMS Annual Meeting, 2026

My presentation is currently scheduled as follows:

Session: Service Operations, Supply Chains, and Experimentation

Day: Monday, November 2

Time: 1:15 PM-2:30 PM

Room: Moscone South-205 (Level 2)

LANGUAGES

Persian: Native | English: Fluent

REFERENCES

Professor Achal Bassamboo

Professor of Operations, Kellogg School of Management, Northwestern University

Email: [a-bassamboo@kellogg.northwestern.edu]

Professor Seyed MR Iravani

Professor, Industrial Engineering and Management Sciences, Northwestern University

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Professor Martin Lariviere

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Professor Izzy Grosz

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